

Notes: Crawford, Pavanini, and Schivardi (AER, 2018)

Asymmetric Information and Imperfect Competition in Lending Markets

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1 Big picture

- **Question.** How do **asymmetric information** and **imperfect competition** interact in lending markets? In particular, when borrower risk is privately known, does stronger bank market power worsen or mitigate the effect of adverse selection on prices, credit supply, and default?
- **Setting.** The paper studies the Italian market for **small business lines of credit** using a matched panel of firms and banks from **1988–1998**. The product is a short-term, uncollateralized line of credit, and the paper focuses on each firm’s **main credit line in the first year it borrows**.
- **Core challenge.** Adverse selection is hard to observe because it is, by definition, based on private information. At the same time, standard empirical work often assumes **perfect competition** in lending, even though local Italian banking markets were concentrated and bank competition plausibly mattered for pricing.
- **Main methodological contribution.** The paper builds and estimates a **structural model** of:
 - firms’ bank choice (credit demand),
 - loan use conditional on borrowing,
 - default conditional on borrowing,
 - bank pricing under Bertrand-Nash competition.

The model allows the unobserved determinants of demand, loan use, and default to be correlated, which lets the authors measure **adverse selection** directly.

- **Why it matters.** In competitive markets, more adverse selection usually raises prices because the average borrower becomes riskier. But with market power, a bank also cares about marginal borrowers. This means competition and asymmetric information may interact in nontrivial ways, which matters for antitrust, prudential regulation, and crisis policy.
- **Main empirical question.** How much adverse selection is present in Italian small-business lending, and how does bank market power change the response of prices, demand, loan use, and default to:
 - more adverse selection,
 - higher bank funding costs,
 - bank mergers?
- **Main empirical results.**
 - The estimated correlation between the unobserved determinants of **demand and default** is **0.16**.
 - The estimated correlation between the unobserved determinants of **loan use and default** is **0.14**.
 - The effect of interest rates on default is positive, which the paper interprets as evidence of **moral hazard**.
 - Doubling adverse selection raises average prices by **1.87 percentage points** (about **12.9%**), lowers demand probabilities by **1.28 percentage points**, slightly lowers loan use, and raises default probabilities by **5.84 percentage points**.

- A one-standard-deviation increase in a bank’s baseline effective markup reduces that adverse-selection-induced price increase by about **6 percentage points**.
- A **5% increase in marginal costs** raises average prices by **1.78 percentage points** and default by **1.84 percentage points**; again, banks with higher market power pass through less.
- A merger between the two largest banks in each local market raises average prices by only **0.15 percentage points** under baseline adverse selection, but under high adverse selection average prices become slightly **negative on a percentage-point basis** and more than **33%** of prices fall.
- **Economic takeaway.** Competition and asymmetric information cannot be studied separately. More competition is not always unambiguously beneficial in credit markets because stronger competition can make banks respond more aggressively to adverse selection and funding shocks.

2 Data and measurement (key objects)

2.1 Main data sources

- **Central Credit Register (Centrale dei Rischi).** Provides loan-level data on individual credit relationships between firms and banks, including the amount granted, the amount used, and—for about **94 large banks**—the interest rate charged.
- **Centrale dei Bilanci (CB).** Provides firm balance-sheet and income-statement data. This dataset also includes the firm’s **Score**, an observable measure of default risk based on financial ratios.
- **Banking Supervision Register.** Provides bank balance-sheet and income-statement information.
- **Local branch data.** Provide information on branch locations and years of presence in local markets.

2.2 Loan, firm, bank, and market definitions

- **Loan product.** The paper focuses on **short-term lines of credit**. These are useful because:
 - the bank can change the interest rate at any time,
 - the contracts are relatively homogeneous,
 - they are not collateralized,
 - they account for more than half of firms’ bank lending in the data.
- **Interest rate.** Defined as the ratio of the payment made in a year by the firm to the bank to the average amount of the loan used.
- **Main credit line.** Since firms often borrow from multiple banks, the paper defines the **main credit line** as the one on which the firm borrows the most. On average, this represents about **75%** of total credit used.

- **Sample restriction.** The paper keeps only the **first year** in which a firm is observed borrowing, dropping 1988 to avoid left-censoring. This leaves **36,520 firms**.
- **Local market.** A market is an Italian **province-year**. Provinces are used because small-business lending is local, because this was the relevant unit in Italian regulation and antitrust practice, and because the average borrower is very close to the nearest branch of its main bank.

2.3 Key descriptive facts from the data

- **Loan-level moments.**
 - **69%** of firms in the sample take up a loan.
 - Conditional on borrowing, firms use on average **67%** of the granted amount.
 - The average amount granted is about **EUR 367,000**.
 - The average interest rate is about **14.24%**.
 - **6%** of main lines default within the paper's default window.
- **Bank-level moments.**
 - The 94 banks that report interest rates account for about **80%** of the lending market.
 - Average total assets are about **EUR 10.7 billion**.
 - The average bank operates in **34 provinces** out of 95.
- **Firm-level moments.**
 - Borrowing firms are larger and more profitable than non-borrowers.
 - Borrowing firms have on average **2.87 lenders**.
 - The average distance to the nearest branch of the main bank is **2.92 km**.
 - The average Score among borrowers is **5.36**.
- **Market-level moments.**
 - There are on average **8.6 banks per province-year**.
 - Each bank has about **15 branches per province-year** on average.
 - The average bank market share is about **8%**.

2.4 Default definition and institutional meaning

- A loan is classified as defaulted if the firm's main line is reported as **bad debt** within **three years** of origination.
- A bad-debt warning is not just one missed payment. It reflects a negative bank assessment of the firm's overall condition and usually precedes formal bankruptcy.
- Default in this environment has a large dynamic consequence: once a bank reports bad debt, a **domino effect** often shuts the firm out of other bank credit as well.
- In the data, **82.3%** of firms receiving such a warning cease borrowing in the same year, **15.1%** in the following year, and all remaining firms within four years.

2.5 Preliminary evidence before the structural model

- Higher local concentration, measured by the **HHI** or the **three-bank concentration ratio**, is associated with higher interest rates.
- Reduced-form positive-correlation tests show that:
 - firms with a higher unexplained propensity to demand credit are more likely to default,
 - firms with a higher unexplained propensity to use more of their granted loan are more likely to default.

These reduced-form patterns are suggestive of asymmetric information and motivate the structural model.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic environment and key assumptions

- In each market-year, firms seek financing for a project and choose one bank for their **main line of credit**, if any.
- Banks compete in **Bertrand-Nash interest rates**.
- Borrowers know the riskiness of their own project, but banks know only the distribution of risk types conditional on observables.
- Banks are assumed to use **interest rates as the only screening device**. The amount granted is taken as exogenous to the pricing decision. This is motivated by the lack of contract exclusivity: firms can borrow from multiple banks, so convex loan menus are hard to implement.

3.2 Demand for the main credit line

The firm's latent utility from borrowing from bank j is

$$U_{ijmt}^D = \bar{\alpha}_0^D + X_{jmt}^D \beta^D + \xi_{jmt}^D + \alpha^D P_{ijmt} + Y_{ijmt}^D \eta^D + \varepsilon_i^D + \nu_{ijmt}^D. \quad (1)$$

The outside option is not to borrow:

$$U_{i0mt}^D = \nu_{i0mt}.$$

Interpretation.

- X_{jmt}^D collects bank-market-year demand shifters such as branch density and years of presence.
- Y_{ijmt}^D collects firm-bank controls.
- P_{ijmt} is the interest rate charged by bank j to firm i .
- ε_i^D is the firm's bank-invariant latent taste for credit. Because firms observe it but banks do not, it is part of the paper's **private information**.
- ν_{ijmt}^D is an idiosyncratic bank-specific demand shock.

The paper uses a mixed-logit bank-choice model. After substituting the predicted-price structure, the choice probability can be written schematically as

$$\Pr_{ijmt}^D = \int \frac{\exp(\tilde{\delta}_{jmt}^D + V_{ijmt}^D)}{1 + \sum_{\ell} \exp(\tilde{\delta}_{\ell mt}^D + V_{i\ell mt}^D)} f(\varepsilon_i^D) d\varepsilon_i^D. \quad (12)$$

Interpretation. The bank-choice model is like a differentiated-products demand system in which banks are differentiated by observable features and latent borrower-specific tastes.

3.3 Loan use and default

Conditional on borrowing, the firm chooses loan use according to the latent index

$$U_{ijmt}^L = \alpha_0^L + X_{jmt}^L \beta^L + \alpha^L P_{ijmt} + Y_{ijmt}^L \eta^L + \varepsilon_i^L. \quad (2)$$

Conditional on borrowing, the firm defaults if the latent utility from default is positive:

$$U_{ijmt}^F = \alpha_0^F + X_{jmt}^F \beta^F + \alpha^F P_{ijmt} + Y_{ijmt}^F \eta^F + \varepsilon_i^F. \quad (3)$$

The key unobservables follow a multivariate normal distribution:

$$\begin{pmatrix} \varepsilon_i^D \\ \varepsilon_i^L \\ \varepsilon_i^F \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_D^2 & \rho_{DL}\sigma_D\sigma_L & \rho_{DF}\sigma_D \\ \rho_{DL}\sigma_D\sigma_L & \sigma_L^2 & \rho_{LF}\sigma_L \\ \rho_{DF}\sigma_D & \rho_{LF}\sigma_L & 1 \end{pmatrix} \right). \quad (4)$$

Interpretation.

- $\rho_{DF} > 0$ means firms with unexpectedly high demand for credit are also unexpectedly more likely to default. This is the paper's first measure of **adverse selection**.
- $\rho_{LF} > 0$ means firms that use unexpectedly more credit are also unexpectedly more likely to default. This is a second measure of **adverse selection**.
- ρ_{DL} allows demand and loan use unobservables to be correlated, but does not by itself identify adverse selection.
- A positive coefficient on price in the default equation, $\alpha^F > 0$, is interpreted as **moral hazard**: exogenously higher interest rates make repayment less likely.

3.4 Bank pricing under imperfect competition

Bank j 's expected profit from lending to firm i is

$$\Pi_{ijmt} = P_{ijmt} Q_{ijmt} (1 - F_{ijmt}) - MC_{ijmt} Q_{ijmt}. \quad (5)$$

Here:

- Q_{ijmt} is expected quantity lent, equal to demand probability times expected loan use,

- F_{ijmt} is expected default probability,
- MC_{ijmt} is the bank's marginal cost of lending to firm i .

The Bertrand-Nash first-order condition gives

$$P_{ijmt} = \frac{MC_{ijmt}}{1 - F_{ijmt} + F'_{ijmt}m_{ijmt}} + \frac{(1 - F_{ijmt})m_{ijmt}}{1 - F_{ijmt} + F'_{ijmt}m_{ijmt}}, \quad (6)$$

where the conventional markup is

$$m_{ijmt} \equiv -\frac{Q_{ijmt}}{Q'_{ijmt}}.$$

Interpretation.

- The first term is the bank's **effective marginal cost**.
- The second term is the bank's **effective markup**.
- Unlike a standard pricing equation, repayment risk enters both pieces because the bank only gets revenue when the loan is repaid.

This pricing equation is the key to the paper's economics. The authors describe two competing channels.

Average borrower effect.

- More adverse selection raises the average default rate of the borrowers who choose to take loans.
- This lowers repayment probabilities and tends to **raise prices**.

Markup-mediated marginal borrower effect.

- More adverse selection also makes default more sensitive to price at the margin.
- If banks have substantial markups, they value marginal borrowers more and may **lower prices** or increase them less in order to keep relatively safer marginal customers.

Bottom line. In competitive markets the first effect dominates, but with enough market power the second effect can offset or even reverse it.

3.5 Price prediction for unchosen banks

A major data problem is that the econometrician only observes prices for firm-bank pairs that actually transact. The paper therefore predicts prices for all feasible bank choices using

$$P_{ijmt} = \gamma_0 + \gamma_1 d_{ijmt} + \gamma_2 \ell_{ijmt} + \lambda_{jmt} + \omega_i^P + \tau_{ijmt}. \quad (7)$$

Interpretation.

- d_{ijmt} is distance from the firm to the closest branch of bank j .
- ℓ_{ijmt} are dummies for the amount granted.
- λ_{jmt} are bank-market-year effects.

- ω_i^P is a firm fixed effect, which captures any borrower characteristic observed by banks but not by the econometrician and common across banks.

This equation is estimated using **all firm-bank relationships** in the first year a firm borrows, not just the main line. The heavy bunching of loan sizes at round amounts—especially **EUR 50,000**, **EUR 100,000**, and **EUR 200,000**—motivates the use of amount dummies rather than a simple linear amount term.

For non-borrowing firms, prices are imputed using propensity-score matching to similar borrowing firms.

3.6 Econometric specification of demand, loan use, and default

The structural demand equation uses the predicted price decomposition and a proxy for soft information based on the pricing fixed effect. This leads to a mixed-logit demand system estimated in two steps. In the first step, the paper uses maximum simulated likelihood and a Berry-Levinsohn-Pakes-style contraction to recover mean utilities $\tilde{\delta}_{jmt}^D$. In the second step, it estimates

$$\tilde{\delta}_{jmt}^D = \bar{\alpha}_0^D + \alpha^D \tilde{P}_{jmt} + X_{jmt}^{D'} \beta^D + \xi_{jmt}^D. \quad (17)$$

Interpretation. This second-stage IV regression is where the demand price coefficient is identified.

Loan use and default are estimated in the first stage using the actual observed interest rate together with the conditional normal structure implied by equation (4). The paper addresses price endogeneity there with a **control-function** approach based on a first-stage pricing residual.

3.7 Recovered effective marginal costs

Once prices, markups, and default probabilities are estimated, the bank's borrower-specific marginal cost is recovered from the pricing equation as

$$\widehat{\text{MC}}_{ijmt} = \tilde{P}_{ijmt} \left[1 - \hat{F}_{ijmt} + \hat{F}'_{ijmt} \hat{m}_{ijmt} \right] - \frac{(1 - \hat{F}_{ijmt}) \hat{m}_{ijmt}}{1 - \hat{F}_{ijmt} + \hat{F}'_{ijmt} \hat{m}_{ijmt}}. \quad (19)$$

Interpretation. This lets the authors back out effective markups directly from the model and then use them as a bank-specific measure of market power in the counterfactual exercises.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive approach

- **Missing prices.** We only observe prices for loans that were actually taken. Demand estimation needs prices for the non-chosen banks too.
- **Soft information.** Banks may observe borrower characteristics not observed by the econometrician, such as reputation or branch-specific knowledge.
- **Price endogeneity.** Banks may charge higher interest rates to borrowers who are unobservably riskier or more dependent on external finance.

4.2 Key identification ideas

- **Multiple-bank borrowing helps identify prices.** Because firms often borrow from several banks, the paper can estimate a pricing regression with **firm fixed effects**. Those fixed effects capture borrower-specific soft information common across banks.
- **Residual-default test validates the price prediction model.** If pricing residuals still predict later default, then the pricing model is missing risk-relevant information used by banks. In specifications without firm fixed effects they do predict default; with firm fixed effects they do **not**. This is a central credibility check.
- **Demand identifies bank choice and price sensitivity.** Cross-bank substitution, together with bank characteristics and predicted prices, identifies the mixed-logit demand system.
- **Loan use and default identify selection and moral hazard.** Observing how borrowing, loan use, and default move together allows the paper to separate:
 - adverse selection through ρ_{DF} and ρ_{LF} ,
 - moral hazard through the direct effect of price on default.

4.3 Instruments

- **Demand equation instruments.** The second-stage demand IV uses bank-region-year measures from the deposit market:
 - deposit interest rates,
 - the value of deposits,
 - the number of deposit accounts.

The logic is that these shift banks' funding conditions but are determined in a different market with different buyers and products.

- **Loan use and default instruments.** For the loan-use and default equations, the paper uses **prices charged by the same bank in other markets** as instruments, combined with bank and market-year fixed effects.
- **Control function.** The residual from the pricing regression is added directly to the loan-use and default equations so that remaining price variation comes from the instrument rather than omitted borrower risk.

4.4 What makes the identification credible

- Large Italian banks relied heavily on **hard information** and scoring systems, according to institutional evidence summarized from a Bank of Italy survey.
- The focus on the **first year** of a firm's main credit line reduces the importance of relationship-specific soft information and dynamic pricing.
- The final pricing regression explains about **71.7%** of price variation, which is high for this literature.

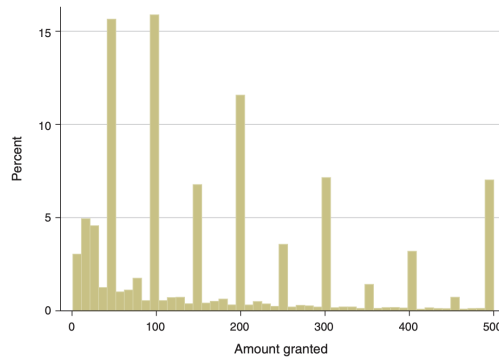


FIGURE 1. DISTRIBUTION OF AMOUNT GRANTED

Notes: Amount granted is in thousands of euros. The observations above €500,000 (15 percent of sample) have been excluded to simplify the interpretation of the graph.

- In the preferred pricing specification, the residual has essentially **no explanatory power for default**, which strongly suggests that the model captures the information used by banks for pricing risk.

4.5 What remains outside the model

- The model abstracts from the dynamics of lending relationships.
- It focuses on the main line rather than the full network of a firm's simultaneous credit relationships.
- It does not let the extent of asymmetric information vary over the business cycle.
- It does not estimate firms' project payoffs directly, so it cannot compute full social welfare.

5 Tables and figures

5.1 Figure 1: distribution of amount granted

Read.

- The distribution of granted loan amounts is heavily bunched at round values.
- More than **40%** of loans are exactly **EUR 50,000**, **EUR 100,000**, or **EUR 200,000**, and about **71%** are exact multiples of **EUR 50,000**.
- This is why the price-prediction equation uses **loan-amount dummies** rather than a single linear term.

5.2 Table 1: summary statistics and market structure

Read.

- The paper starts from **36,520** firms.

TABLE 1—SUMMARY STATISTICS

Variable	Observations	Mean	SD	Observations	Mean	SD
<i>Panel A. Loan level</i>						
Demand	36,520	0.69	0.46			
Loan use	25,351	246.1	444.8			
Default	25,351	0.06	0.23			
Amount granted	25,351	367.3	476.7			
Interest rate	25,351	14.24	4.58			
<i>Panel B. Bank level</i>						
Total assets	900	10,727	16,966			
Employees	896	3,180	4,583			
Number of provinces	861	34.54	30.19			
<i>Panel C. Firm level</i>						
	Borrowing firms			Non-borrowing firms		
Total assets	25,351	11,336	19,825	11,169	3,622	8,601
Net assets	25,351	2,300	6,499	11,169	931	3,621
Intangible assets	25,351	360	2,066	11,169	131	1,079
Intangible/total assets	25,351	0.16	0.23	11,169	0.22	0.29
Profits	25,351	1,033	2,910	11,169	292	1,403
Cash flow	25,351	673	2,185	11,169	255	1,183
Sales	25,351	14,478	25,014	11,169	5,029	12,084
Trade debit	25,351	1,710	3,546	11,169	814	3,617
Short-term debt	25,351	2,263	5,697	11,169	125	1,512
Leverage	25,351	0.55	0.86	11,169	0.21	0.70
Firm's age	25,351	12.92	12.89	11,169	10.73	12.12
Score	25,351	5.36	1.77	11,169	4.78	2.14
Distance to branch (km)	25,351	2.92	6.72			
Number of lenders	25,351	2.87	2.22			
Share of main line	19,751	0.76	0.25			

<i>Panel D. Market level</i>			
Number of banks	702	8.60	4.83
Number of branches	6,036	14.95	24.46
Share of branches	6,036	0.06	0.08
Years in market	6,036	21.06	14.23
Market shares	6,036	0.08	0.08
Deposit amount	2,566	21,113	18,883
Number deposit accounts	2,566	654	523
Deposit interest rate	2,566	6.51	1.72

Notes: This table reports sample statistics for the variables in our analysis. In panel A, an observation is a firm for the first variable and a loan for the others. Demand is a dummy variable indicating whether a firm obtained a credit line. Loan use is the amount of loan used in thousands of euros. Default is a dummy for a firm having any of its loans classified as bad within the next three years (see Section IC for further details). (Loan) amount granted is in thousands of euros. Interest rate is a percentage. In panel B, an observation is a bank-year. Total assets are in thousands of euros. Employees is the number of employees at the end of the year. Number of provinces is the number of provinces where a bank is actively lending. In panel C, an observation is a firm. Total, net, and intangible assets, profits, cash flow, sales, trade debit, and short-term debt are in thousands of euros. Intangible/total assets is the ratio of intangible over total assets. Leverage is the ratio of debt over equity. Firm's age measures the years since a firm's foundation. Score is an indicator of the risk of the firm computed each year by the CB (higher values indicate riskier companies, see Section IB for more details). Distance to branch is the distance in kilometers between the city council of each firm and the city council of the closest branch of the bank it borrows from, calculated using the geographic coordinates. Number of lenders is the number of banks from which a firm opens a line of credit. Share of main line represents the ratio of credit used from a firm's main line of credit over total credit used, when credit used is positive. In panel D, an observation is province-year for the number of banks, bank-province-year for the subsequent four variables, and bank-region-year for the last three variables. Number and share of branches are per bank-province-year. Years in market are the number of years a bank has been in a province since 1959. Market shares are in terms of number of borrowers. Deposit amount is the total value of a bank's deposits in a region-year in thousands of euros. Number of deposit accounts is in thousands. Deposit interest rate is a percentage.

- **69%** borrow; conditional on borrowing, average loan use is about **EUR 246,000** and average interest rates are **14.24%**.
- Default on new main lines is **6%**.
- Borrowing firms use multiple lenders on average (**2.87 lenders**), which is important for both the institutional interpretation and the price-prediction strategy.
- Local markets are concentrated but not monopolistic: there are on average **8.6 banks per province-year**.

5.3 Table 2 and Table 3: price prediction and the role of firm fixed effects

Read.

- Table 2 shows that the preferred pricing regression is the one with **firm fixed effects**. It reaches an R^2 of about **0.717**, much higher than the simpler specifications.
- The negative coefficients on larger granted-amount dummies indicate that bigger credit lines tend to have lower interest rates.
- Distance to the closest branch also matters positively for price.
- Table 3 is the crucial validation exercise. In columns without firm fixed effects, the pricing residual significantly predicts default. In the preferred column with firm fixed effects, the residual coefficient is about **0.04** and statistically insignificant.
- Interpretation: the firm fixed effects appear to capture the borrower information used by banks in pricing, including soft information that would otherwise contaminate the adverse-selection estimates.

5.4 Table 4: structural estimates

Read.

TABLE 2—PRICE REGRESSIONS

Variables	(1)	(2)	(3)	(4)
Amount granted	−2.37 (0.07)	−2.39 (0.07)	—	—
50,001–100,000	—	—	−1.47 (0.07)	−0.92 (0.09)
100,001–150,000	—	—	−2.44 (0.08)	−1.55 (0.09)
150,001–200,000	—	—	−2.77 (0.10)	−1.98 (0.10)
200,001–300,000	—	—	−3.18 (0.10)	−2.19 (0.10)
300,001–400,000	—	—	−3.72 (0.11)	−2.63 (0.10)
400,001–500,000	—	—	−3.99 (0.12)	−2.88 (0.11)
500,001–1,000,000	—	—	−4.37 (0.12)	−3.06 (0.10)
1,000,001–3,000,000	—	—	−5.02 (0.13)	−3.44 (0.12)
Distance to branch	−0.94 (0.21)	−0.70 (0.20)	−1.33 (0.20)	−0.40 (0.30)
Constant	16.80 (0.18)	15.53 (0.13)	17.52 (0.15)	15.49 (0.81)
Firm controls	Yes	Yes	Yes	No
Bank fixed effects	Yes	No	No	No
Area fixed effects	Yes	No	No	No
Year fixed effects	Yes	No	No	No
Bank-area-year fixed effects	No	Yes	Yes	Yes
Firm fixed effects	No	No	No	Yes
R ²	0.294	0.319	0.365	0.717
Observations	92,596	92,596	92,596	92,602

Notes: The table shows the results of OLS regressions of the interest rate (in percentage points) on a series of controls and dummies. An observation is a firm-bank. The sample only includes the first year in which a firm borrows, excluding 1988. See Table 1 for variables' definitions. Firm controls include sector and score fixed effects, sales, total assets, net assets, profits, cashflow, leverage, and short-term debt. Firm controls and distance are rescaled to interpret the coefficients more easily: the linear term for amount granted is measured in units of £10,000, and distance to branch is in 100 km. Sector fixed effects group sectors into 3 categories: *primary sectors* (primary goods, minerals extraction, chemicals, metals, and energy), *manufacturing and construction*, and *commerce and services*. Area fixed effects are based on four geographic areas of similar size in terms of population: North-West, North-East, Center, and South. Standard errors are clustered at the bank-area-year level in columns 1–3, and at the firm level in column 4.

TABLE 3—THE ABILITY OF PRICING RESIDUALS TO PREDICT DEFAULT

Variables	(1)	(2)	(3)	(4)
Residual	0.26 (0.03)	0.27 (0.03)	0.26 (0.03)	0.04 (0.05)
Residual SD	0.04	0.04	0.04	0.02
Default SD	0.24	0.24	0.24	0.24
1 residual SD versus percent of 1 default SD	4.7	4.7	4.4	0.3
Firm controls	Yes	Yes	Yes	Yes
Bank fixed effects	Yes	No	No	No
Area fixed effects	Yes	No	No	No
Year fixed effects	Yes	No	No	No
Bank-area-year fixed effects	No	Yes	Yes	Yes
Amount granted fixed effects	No	No	Yes	Yes
R ²	0.088	0.123	0.123	0.121
Observations	35,316	35,316	35,316	35,316

Notes: The table shows the results of OLS regressions of a dummy equal to 1 if the firm defaults on a loan, and 0 otherwise, on the residuals of the pricing regressions reported in Table 2 and other controls. An observation is a firm. All the controls used are the same as in each respective column in Table 2, with the exception of column 4 where we cannot include firm fixed effects. Residual is divided by 100 for ease of interpretation. See Table 1 for variables' definitions. Standard errors are clustered at the bank-year-province level.

- The demand price coefficient is negative, as expected, so higher interest rates reduce a firm's probability of choosing a bank.
- The default price coefficient is positive, which the paper interprets as **moral hazard**.
- The central results are the correlation estimates:
 - $\rho_{DF} = \mathbf{0.16}$,
 - $\rho_{LF} = \mathbf{0.14}$.

Both are highly statistically significant and imply adverse selection on both the extensive and intensive borrowing margins.

- The paper also reports that a **10% increase in a bank's interest rate** reduces its own market share by slightly more than **10%** and increases competitors' shares by slightly less than **1%**.

	Demand	Loan use	Default
<i>Firm level</i>			
Price			
Interest rate	−1.45 (0.62)	−0.01 (0.00)	1.06 (0.02)
Assets			
Total assets	5.84 (0.08)	0.09 (0.00)	−0.04 (0.03)
Intangible/total assets	−0.82 (0.05)	−0.01 (0.01)	0.08 (0.05)
Profitability			
Profits	1.12 (0.05)	0.01 (0.00)	0.03 (0.02)
Cash flow	−0.93 (0.05)	−0.05 (0.00)	−0.12 (0.02)
Sales	7.16 (0.07)	−0.01 (0.00)	−0.34 (0.04)
Debt			
Trade debit	−3.44 (0.05)	−0.04 (0.00)	0.12 (0.02)
Others			
Firm's age	0.23 (0.03)	0.00 (0.00)	0.02 (0.03)
Distance to branch	−1.22 (0.03)	−0.01 (0.01)	−0.04 (0.06)
<i>Bank level</i>			
Number of branches	4.38 (0.15)	0.01 (0.00)	0.03 (0.02)
Share of branches	0.53 (0.04)	−0.07 (0.01)	−0.27 (0.10)
Years in market	0.06 (0.03)	0.01 (0.00)	−0.13 (0.02)
<i>First-stage fixed effects</i>			
Score	Yes	Yes	Yes
Sector	Yes	Yes	Yes
Loan amount	Yes	Yes	Yes
Bank-market-year	Yes	No	No
Bank	No	Yes	Yes
Market-year	No	Yes	Yes
Observations	506,230	25,351	25,351
<i>Covariance matrix (Σ)</i>			
	$\sigma_D = 0.34$ (0.00)		
	$\rho_{DL} = 0.10$ (0.00)	$\sigma_L = 0.30$ (0.00)	
Adverse selection	$\rho_{DF} = 0.16$ (0.00)	$\rho_{LF} = 0.14$ (0.00)	$\sigma_F = 1$

Notes: All coefficients are estimated in the first stage, with the exception of the Interest rate, the number of branches, the share of branches, and the years in market for the demand equation, that are estimated in the second stage. Second-stage fixed effects, only for the demand equation, are at the bank, market, and year level. See Table 1 for variables' definition. Standard errors are in brackets. First-stage standard errors are calculated by the inverse of the Information matrix, obtained providing the solver with analytical gradient and Hessian. Second-stage standard errors are computed with 200 bootstrap replications.

5.5 Table 5: fit of the model and recovered markups

Read.

- The model matches means well:
 - actual demand mean = predicted demand mean = **7.21**,
 - actual loan use = **246.06** vs. predicted = **245.42**,
 - actual default = **5.83** vs. predicted = **5.82**.
- Predicted prices average **14.48%**.
- Predicted effective marginal costs average **13.77%**.
- Predicted effective markups average about **0.71 percentage points**, roughly **5%** of the average interest rate.

TABLE 5—DESCRIPTIVES ON MODEL FIT

Variables	Observations	Mean	SD
Actual demand	506,230	7.21	25.87
Predicted demand	506,230	7.21	15.16
Actual loan use	25,351	246.06	444.75
Predicted loan use	25,351	245.42	297.34
Actual default	25,351	5.83	23.44
Predicted default	25,351	5.82	9.74
Predicted price	452,594	14.48	4.54
Predicted effective marginal cost	452,594	13.77	4.57
Predicted effective markup	452,594	0.71	0.48

Notes: All variables are predicted at the firm level. Actual demand is a dummy equal to 1 if the firm takes the loan from a bank and 0 otherwise, multiplied by 100. Predicted demand is the respective demand probability from our model, again multiplied by 100. Actual and predicted loan use refer to loan use in the data and in our model, in thousands of euros. Actual default is a dummy equal to 1 if the firm defaults and 0 otherwise, multiplied by 100 for ease of interpretation. Predicted default is the respective default probability from our model, again multiplied by 100. Predicted price is the interest rate predicted by our model as of equation (6). Predicted effective marginal cost and predictive effective markup are computed based on the two right-hand-side terms in equation (6), with effective markup being the negative of the second term. Predicted price, effective marginal cost, and effective markup have fewer observations than actual and predicted demand as there is no predicted price for the outside good.

TABLE 6—SUMMARY STATISTICS FOR COUNTERFACTUAL CHANGES IN OUTCOMES FOR HIGHER ADVERSE SELECTION AND HIGHER MARGINAL COSTS

Variables	Observations	Higher adverse selection		Higher marginal costs	
		Mean	SD	Mean	SD
Percentage point price variation	35,563	1.87	5.76	1.78	3.50
Percent price variation	35,563	12.9	40.9	11.0	18.6
Percentage point demand variation	35,563	−1.28	6.17	−1.47	4.96
Percent demand variation	35,563	−4.1	14.5	−5.0	10.0
Loan use variation	35,563	−0.7	2.4	−0.7	1.4
Percent loan use variation	35,563	−0.3	2.0	−0.6	2.2
Percentage point default variation	35,563	5.84	22.37	1.84	4.58

Notes: These variables are average changes at the firm-market-year level. See footnote 45 for variable definitions. Loan use changes are in thousands of euros. Higher adverse selection corresponds to a doubling of the estimated values for ρ_{DF} and ρ_{LF} from Table 4; higher marginal costs corresponds to a 5 percent (approximately 70 basis points) increase in banks' marginal costs.

5.6 Table 6 and Table 7: counterfactuals for more adverse selection and higher capital costs

Read.

- **More adverse selection.** Doubling ρ_{DF} and ρ_{LF} :
 - raises average prices by **1.87 percentage points (12.9%)**,
 - lowers demand by **1.28 percentage points**,
 - slightly lowers loan use,
 - raises default by **5.84 percentage points**.
- **Higher marginal costs.** Raising marginal cost by 5%:
 - raises prices by **1.78 percentage points (11.0%)**,
 - lowers demand by **1.47 percentage points**,
 - lowers loan use slightly,
 - raises default by only **1.84 percentage points**.
- The bigger default response under adverse selection than under higher marginal costs is important: adverse selection worsens the **composition** of borrowers, not just the level of interest rates.
- Table 7 shows that market power mitigates both shocks. A one-standard-deviation increase in effective markup lowers the price response to higher adverse selection by about **6 percentage points** and lowers the pass-through of higher marginal costs by about **6.6 percentage points**.

5.7 Table 8: merger simulation

Read.

- The paper simulates a merger between the **two largest banks** in each local market.
- Under baseline adverse selection:
 - average price rises by **0.15 percentage points (1.0%)**,

TABLE 7—REGRESSIONS OF COUNTERFACTUAL OUTCOMES' CHANGES ON MARKUPS FOR HIGHER ADVERSE SELECTION AND HIGHER MARGINAL COSTS

Variables	ΔP_{jmt}	ΔQ_{jmt}^D	ΔQ_{jmt}^L	ΔF_{jmt}
Higher adverse selection				
Effective markup	-0.38 (0.06)	0.15 (0.02)	0.01 (0.00)	-0.13 (0.02)
Bank-market-year fixed effects	Yes	Yes	Yes	Yes
R^2	0.134	0.155	0.092	0.097
Observations	35,563	35,563	35,563	35,563
Higher marginal costs				
Effective markup	-0.42 (0.05)	0.22 (0.03)	0.01 (0.00)	-0.11 (0.01)
Bank-market-year fixed effects	Yes	Yes	Yes	Yes
R^2	0.433	0.451	0.112	0.447
Observations	35,563	35,563	35,563	35,563

Notes: An observation is a firm-market-year. Price, demand probabilities, and loan use changes are measured in percentages. Default changes are measured in percentage points. See footnote 45 for dependent variables' definition. Effective markup is constructed as the negative of the second term on the right-hand side of equation (6). Standard errors are clustered at the bank-market-year level.

TABLE 8—SUMMARY STATISTICS FOR COUNTERFACTUAL CHANGES IN OUTCOMES FOR MERGER UNDER BASELINE AND HIGH ADVERSE SELECTION

Variables	Observations	Baseline adverse selection			High adverse selection		
		Mean	SD	< 0	Mean	SD	< 0
<i>All banks</i>							
Percentage point price variation	35,563	0.15	1.52	16.4%	-0.02	1.57	33.2%
Percent price variation	35,563	1.0	9.9		0.4	9.3	
Percentage point demand variation	35,563	-0.19	2.67		0.08	2.98	
Percent loan use variation	35,563	-0.1	1.1		-0.1	1.7	
Percentage point default variation	35,563	0.14	1.18		-0.03	0.96	
<i>Merged banks</i>							
Percentage point price variation	35,563	0.73	6.45	17.1%	0.15	7.64	33.1%
Percent price variation	35,563	5.7	47.1		4.5	47.3	
<i>Non-Merged banks</i>							
Percentage point price variation	35,196	0.06	0.95	0%	-0.04	0.13	11.4%
Percent price variation	35,196	0.3	4.4		-0.0	0.4	

Notes: These variables are average changes at the firm-market-year level. See footnote 45 for variables' definition. Loan use changes are in thousands of €. There are less firms in this table compared to Table 6 because in three markets there is only one bank in a market so no merger takes place. Note that in the High adverse selection case the mean of the percent Price variation is positive, but the mean of the percentage point price variation is negative. This is driven by the different skewness of the two distributions.

- average demand and loan use fall,
- default rises,
- only **16.4%** of prices fall.
- For merged banks specifically, prices rise by about **0.73 percentage points (5.7%)** on average.
- Under high adverse selection:
 - the average price effect becomes slightly **negative on a percentage-point basis** (about **-0.02**),
 - demand rises slightly,
 - default falls slightly,
 - more than **33%** of prices fall.
- Interpretation: stronger adverse selection can make banks value marginal borrowers more, so the usual price-increasing effect of higher concentration becomes weaker and can even reverse.

6 Short summary

- **Method.** Estimate a structural model of bank choice, loan use, default, and bank pricing in the Italian market for small-business lines of credit.
- **Identification.** Use multiple-bank borrowing to estimate prices with firm fixed effects, use deposit-market variables as instruments in demand, and use prices in other markets plus a control function for loan use and default.
- **Main findings.** There is statistically significant adverse selection on both the extensive and intensive margins of borrowing, and there is evidence of moral hazard.
- **Key result.** Market power and asymmetric information interact strongly: more competition does not mechanically improve outcomes if it intensifies the adverse-selection problem.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the Italian market for small-business credit is characterized by both **adverse selection** and **imperfect competition**.
- It also shows that these two frictions are not additive or separable: the response of prices, credit, and default to one friction depends on the strength of the other.

7.2 Economic intuition

- In a lending market, unlike in a standard product market, the seller only earns revenue if the customer repays.
- Adverse selection therefore rotates something like the bank's **marginal revenue** curve rather than just raising average cost.
- A competitive bank, with little markup to lose, reacts to worse selection mainly by raising prices.
- A bank with market power has more to gain from keeping relatively safe marginal borrowers, so it may raise prices by less or even cut them.

7.3 Takeaway

This paper is useful because it gives a concrete structural answer to a policy question that is often discussed only qualitatively: **how does competition interact with borrower selection?** The answer is that more competition is not always stabilizing in credit markets. When asymmetric information is strong, market power can cushion some of its effects, which means competition policy and financial-stability policy need to be thought about jointly rather than separately.